IT315: Object-Oriented Programming II

Iris Flower Classification

Classifying Iris Species Using Object-Oriented Programming & ML



Project **Overview**

Classifying Iris flowers into three distinct species using machine learning, based on four key botanical measurements from the classic Fisher dataset.

- Setosa
- Versicolor
- Virginica

based on:

Sepal Width • Sepal Length
Petal Length • Petal Width



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Project Objectives

Our Iris Flower Classification Project combines machine learning techniques with software engineering principles to create a practical, interactive application. These four core objectives guide our development process and demonstrate key concepts .

Build ML Classifier



Apply OOP Concepts



Create Interactive GUI



Visualize Iris Dataset



Model Training **Process**

Dataset divided into 80% training and 20% testing sets. This split ensures sufficient data for learning while reserving samples for unbiased validation.

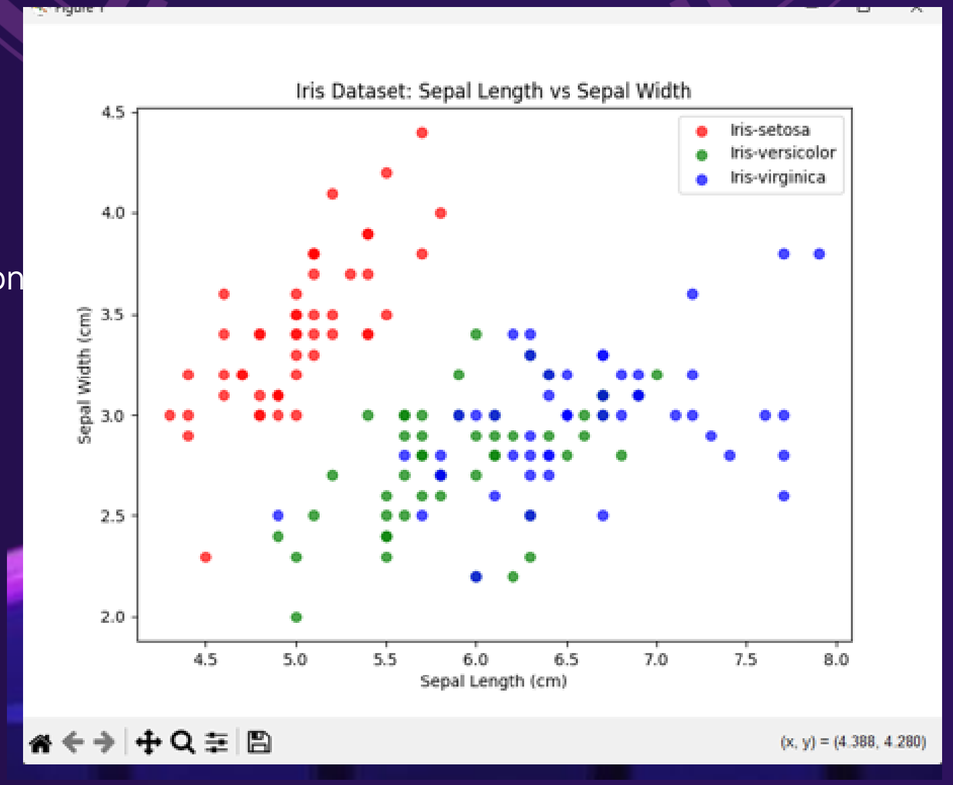
Logistic Regression model trained using Scikit-Learn's fit() method. Model learns decision boundaries from training features to classify Iris species accurately.

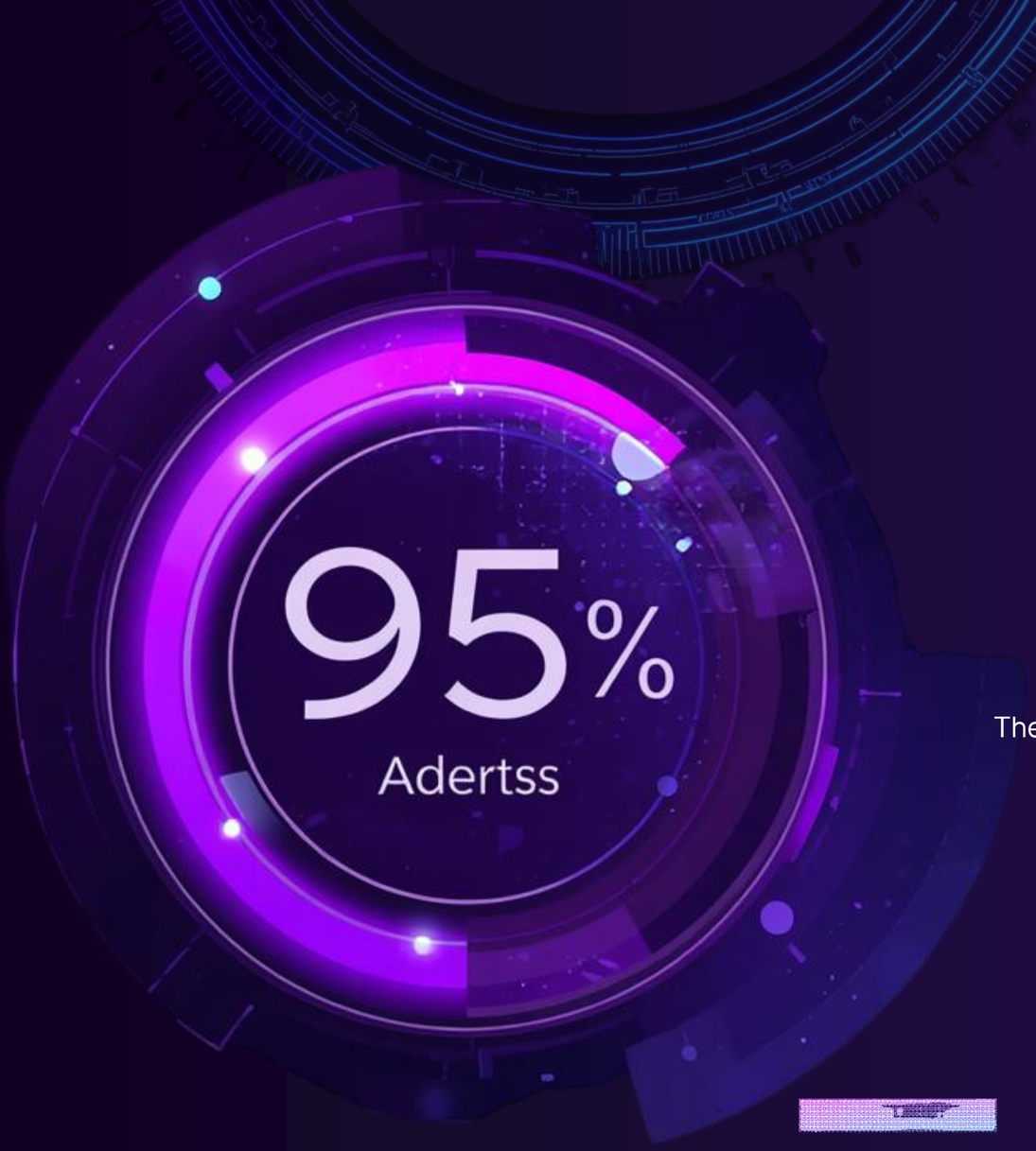
Data Split ***Model Fitting*** ***Testing*** ***Evaluation***

Machine Learning **Model**

Logistic Regression algorithm selected for its proven effectiveness on the Iris classification task, offering the ideal balance of simplicity and performance.

- Simple & Interpretable
- High Accuracy
- Efficient Training
- Fast Prediction
- Proven Results on Iris Dataset





Results and **Accuracy**

Our Logistic Regression model achieved excellent performance metrics on the Iris dataset classification task:

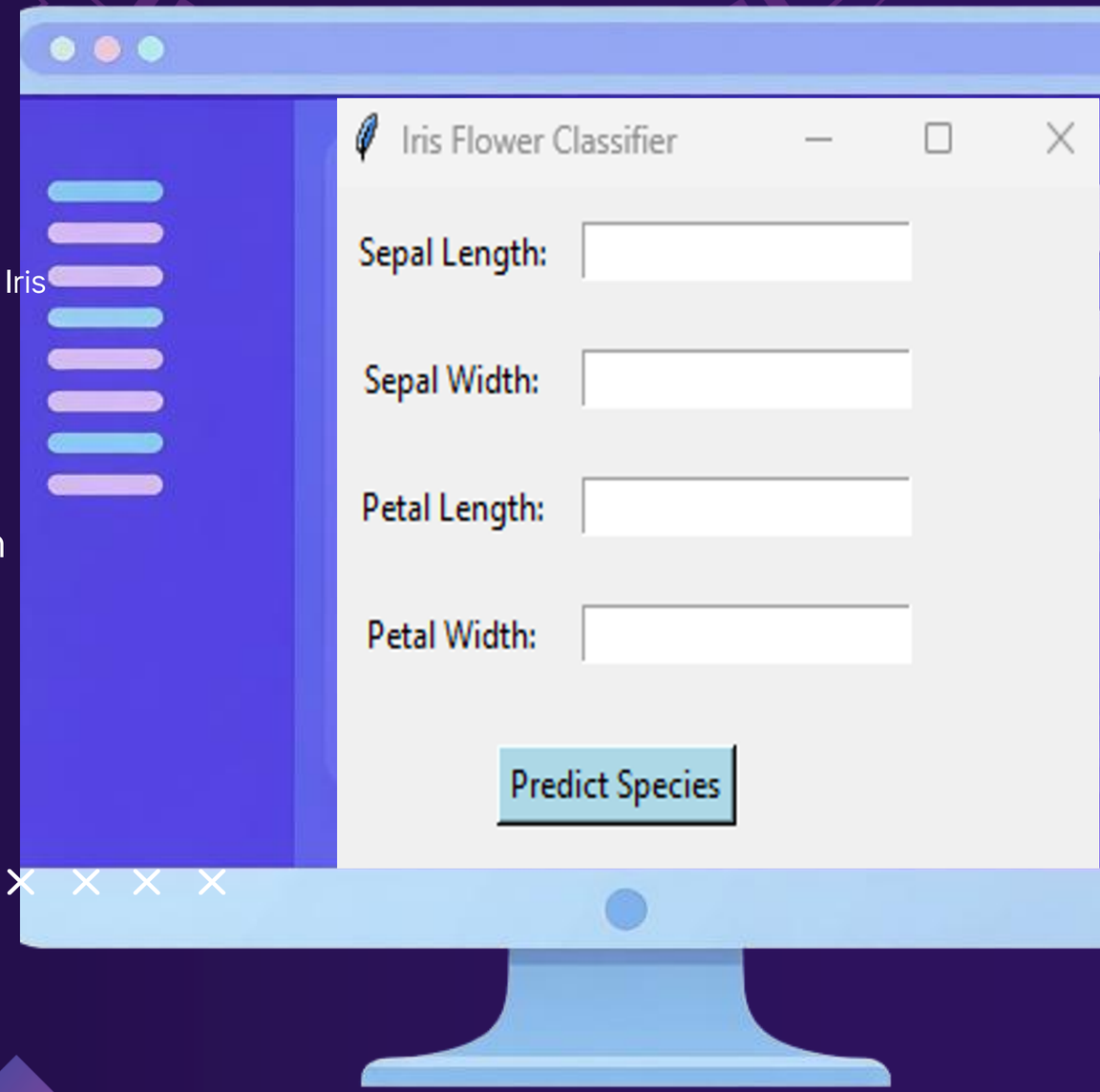
- Prediction Accuracy: ~95%+ on test data
- Fast Classification: Real-time prediction capability
- Consistent Results: Reliable performance across multiple test runs
- Efficient Processing: Minimal computational overhead

The model demonstrates strong generalization, accurately distinguishing between Setosa, Versicolor, and Virginica species based on the four measurement features.

Tkinter
GUI Application

Our interactive desktop application provides a simple, intuitive interface for classifying Iris flowers in real-time using machine learning predictions.

- Input Four Measurements
- Predict Species Button
- Pop-up Result Dialog
- Intuitive Design



Technologies Used

Key tools and libraries powering our Iris Classification project, enabling data manipulation, machine learning, visualization, and user interface development.

Python

Core programming language providing the foundation for all project components with clean, readable syntax.

Pandas

Data manipulation library for loading, cleaning, and exploring the Iris dataset with powerful dataframes.

Scikit-Learn

Machine learning library providing Logistic Regression algorithm for training and prediction.

Matplotlib

Visualization library for creating scatter plots and charts to analyze data patterns.

Tkinter

GUI framework for building the interactive desktop application interface.



Project Conclusion

Successfully classified Iris flowers into three distinct species (Setosa, Versicolor, Virginica) with high accuracy using machine learning techniques.

Combined machine learning algorithms with object-oriented programming principles to create a well-structured, maintainable codebase.

Developed an interactive desktop application with an intuitive Tkinter GUI for real-time flower species prediction.

Demonstrated core software engineering principles including MVC architecture, modular design, and clean code practices.

- ***Accurate Classification***
- ***ML + OOP Integration***
- ***Interactive GUI App***
- ***Engineering Excellence***

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Thank You!

Questions?



Iris Flower Classification Project